Lab Task#7

Sec A

Note: All statements should be commented. Plagiarism will get straight zero.

Starter code is given for both questions. Do not change the parts marked GIVEN; complete every TODO. Add a docstring to every function you write. Submit all your files in one zipped folder.

Question 1

A teacher wants random practice marks for testing and wants to schedule vivas. Half of the program is already written in **Q1.py**:

• `curve(marks)` (given): returns the curved marks, where every mark m becomes ceil(10 × √m).

• `viva_schedule(rolls, start)` (given): prints one viva date per student from the date start, skipping Saturday and Sunday.

• `rolls` (given): the list of 8 roll numbers of the class.

Your task:

1) Complete `generate_marks(count, low, high)` so that it returns a list of count random integers between low and high (both included).

2) Call `random.seed(25)` first, so that everyone gets the same numbers. Generate 10 marks between 35 and 100 and print them.

3) Print the mean and the median of the marks using the statistics module.

4) Using the given `curve()`, print the curved marks and their mean.

5) Select 3 different students from rolls for viva using `random.sample` and print them. Then put them in a random order using `random.shuffle` and print them again.

6) Using the given `viva_schedule()`, print the viva dates of your shuffled students, starting from `date(2026, 10, 16)`.

7) One student from the whole class will help the teacher during the vivas. Pick and print this student using `random.choice`.

Hints:

• `random.randint(a, b)` includes both a and b.

• `random.shuffle(lst)` changes the list itself and returns None, so do not write `x = random.shuffle(lst)`.

• Observe the dates printed by `viva_schedule`: why is there no viva on 17 and 18 Oct 2026?

Question 2

A stationery shop has started organising its billing code as a package. Part of the package is already written:

```
Q2/
├── main.py             (starter: you complete it)
└── shop/
    ├── __init__.py     (given) SHOP_NAME and a welcome message
    ├── catalog.py      (given) PRICES and STOCK dictionaries
    ├── validators.py   (given) is_valid_phone(phone) + test block
    ├── inventory.py    (given) can_sell(item, qty), update_stock(rows)
    └── billing.py      (starter: you complete it)
```

Read the given modules first, then:

1) Complete **billing.py**:

• `make_bill(cart)` returns rows (a list of (item, qty, price, amount) tuples for the items that can be sold), skipped (a list of items that cannot be sold) and the total. Use the given `can_sell()` to decide.

• `apply_discount(total, percent)` returns the discount and the amount to pay.

2) Complete **main.py**: write `process_order(customer, phone, cart)` that

• rejects the order if the phone number is invalid (use the given `is_valid_phone()`) and returns 0;

• otherwise makes the bill and prints the rows and the skipped items;

• applies a 10% discount if the bill is more than 1500 rupees and prints the total, the discount and the amount to pay;

• reduces the stock using the given `update_stock()` and returns the amount paid.

3) Print shop.SHOP_NAME, process all the given orders one after the other, then print the sales of the day and the stock.

4) Run `python shop/validators.py` and then `python main.py`. In a comment in main.py, explain why the test block of validators.py runs only in the first case.

Hints:

• A module of the package imports another module of the same package with the full name, e.g. in billing.py: `from shop.catalog import PRICES` and `from shop.inventory import can_sell`.

• A function can return several values separated by commas. They come back as a tuple, which you unpack just like in Lab 06:

```
def min_max(nums):
    return min(nums), max(nums)


low, high = min_max([4, 9, 2])   # low = 2, high = 9
```

• STOCK is the same dictionary in every module (a module is loaded only once), so the next customer sees the reduced stock. Check Hina's bill: why are notebook and geometry box skipped?

• Run the commands in the terminal from inside the Q2 folder.